Formal Languages and Finite Automata
Homework #3

1. (30%) Given the context-free grammar and w = aaabbbb
   $S \rightarrow aSAB \mid aa$
   $A \rightarrow B$
   $B \rightarrow bbB \mid bb$
   (a) Show the leftmost derivation of w.
   (b) Show the rightmost derivation of w.
   (c) Show the parse tree of w.

2. (10%) Find the context-free grammar for the following language L:
   (a) $L = \{a^n b^m: n \geq 0, m \geq 0\}$.
   (b) $L = \{a^n b^m c^k: n = m + k, n \geq 0, m \geq 0, \text{ and } k \geq 0\}$.

3. (10%) Show that the following grammar that describes the if-then and if-then-else structures is ambiguous:
   $S \rightarrow i\ E\ t\ S \mid i\ E\ t\ S\ e\ S \mid a$
   $E \rightarrow b$

4. (10%) Find an s-grammar for $L = \{a^{2n} b^n: n \geq 1\}$.

5. (20%) For the following grammar, remove all unit-productions, all useless productions, and all λ–productions:
   $S \rightarrow AB \mid C \mid a$
   $A \rightarrow aA \mid B \mid \lambda$
   $B \rightarrow bB \mid b \mid \lambda$
   $C \rightarrow cC \mid D$
   $D \rightarrow d \mid \lambda$
   $E \rightarrow aE \mid F$
   $F \rightarrow b$

6. (20%) Given the following context-free grammar:
   $S \rightarrow aBA \mid abC$
   $A \rightarrow bS \mid aa$
   $B \rightarrow aB \mid b$
   $C \rightarrow BA \mid bb$
   (a) Convert the grammar into Chomsky normal form.
   (b) Convert the grammar into Greibach normal form.